LOW CLUTCH CLEARANCE MEASUREMENT/ADJUSTMENT

id051700664500

Preparation Before Servicing

1. Print out the measurement/adjustment value input sheet.(See MEASUREMENT/ADJUSTMENT VALUE INPUT SHEET.)

Note

- When performing the measurement/adjustment, input the measured and calculated values into the measurement/adjustment value input sheet.
- If the measurement/adjustment value input sheet has already been printed out for the other measurements/adjustments, use the sheet.

Low Clutch Clearance Measurement

1. Assemble the snap ring to the position shown in the figure.

Caution

- **Assemble so that the end gap of the snap ring is positioned diagonally opposed to the end gap of the snap ring for the high clutch.**
- **After assembling the snap ring, verify that the snap ring is securely inserted into the bottom of the snap ring groove.**

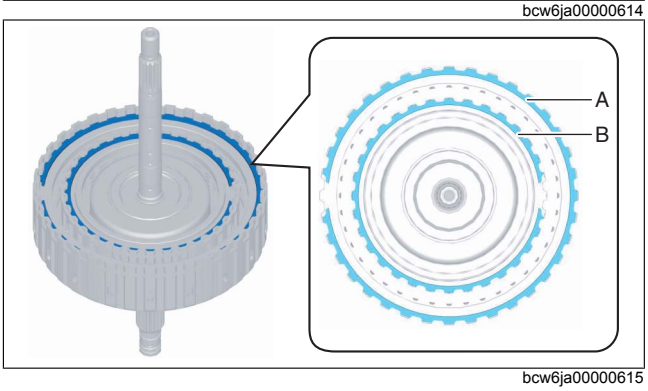
Note

- Snap ring size: Outer diameter approx. 158.8 mm {6.252 in}



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A : Snap ring (low clutch)
B : Snap ring (high clutch)



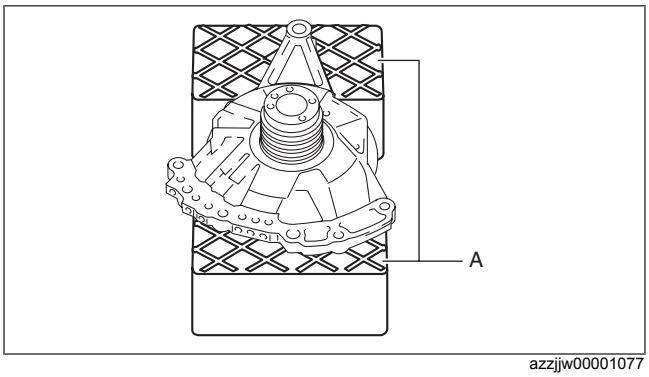
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2. Set the oil pump on the workbench as shown in the figure.

Caution

- **To reduce error during the low clutch clearance measurement, use the rubber plates to adjust the alignment surface of the oil pump with the transaxle case so that it is level.**

A : Rubber plate

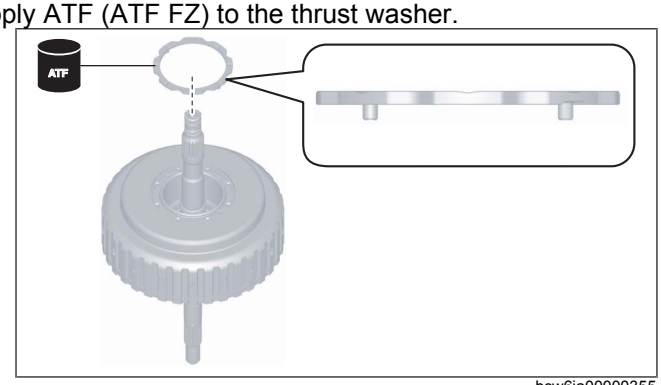


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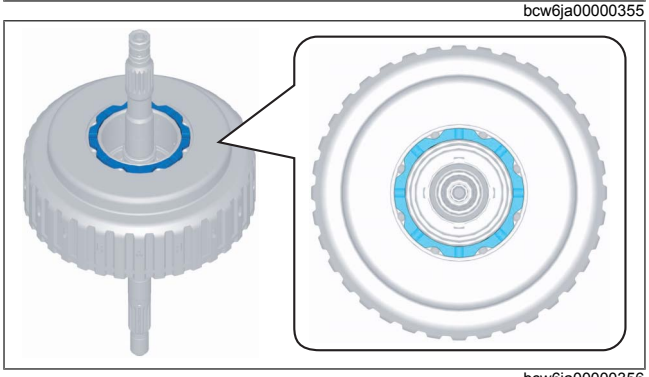
3. Assemble the thrust washer to the clutch component using the following procedure:

Note
- Thrust washer size: Inner diameter approx. 58.7 mm {2.31 in}

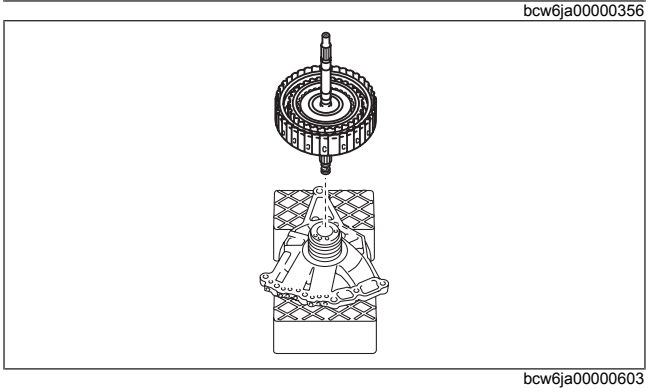
(1) To prevent the thrust washer from dropping out, apply ATF (ATF FZ) to the thrust washer.
(2) Assemble the thrust washer.

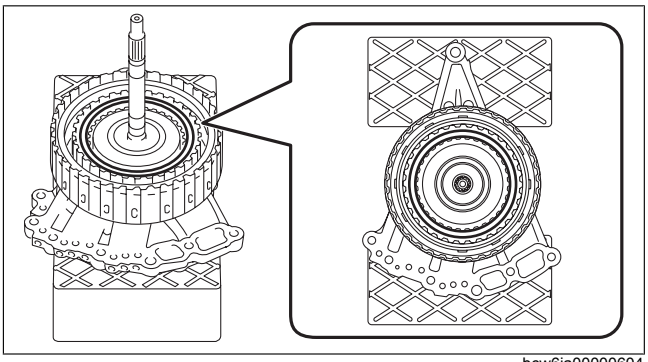


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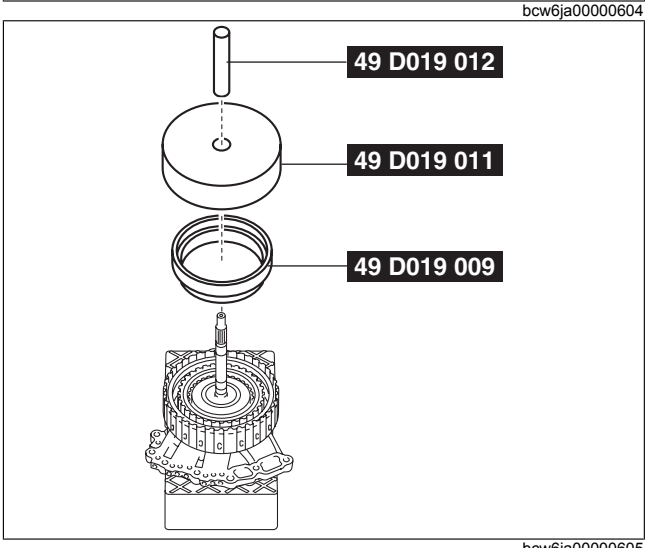

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4. Assemble the parts assembled together in Step 3 to the oil pump.

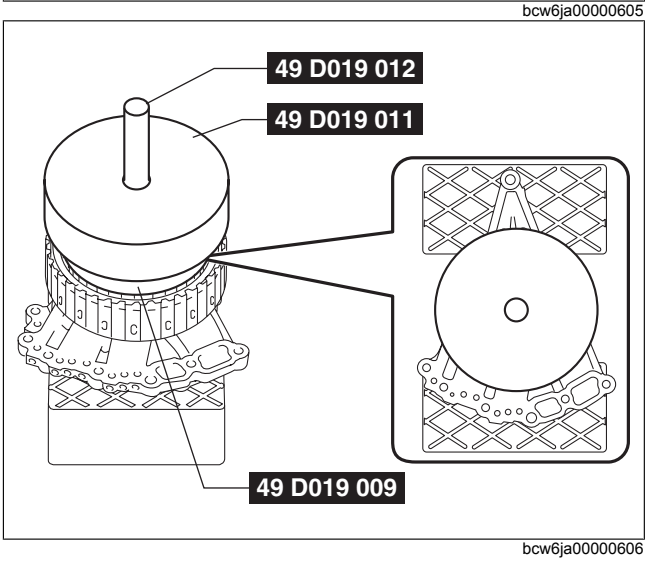

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5. Install the SSTs.



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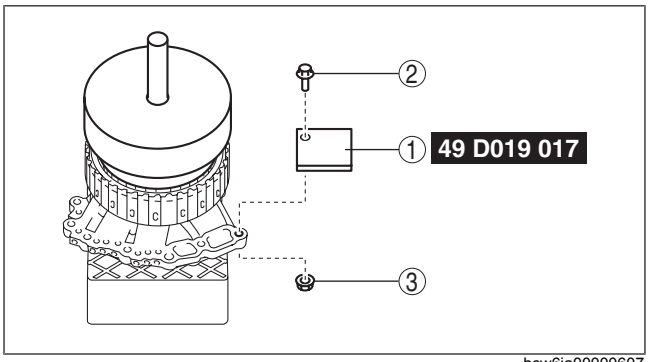


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6. Set the measuring instrument to the oil pump using the following procedure:
   (1) Install the SST for securing the magnetic stand using the procedure shown in the figure.

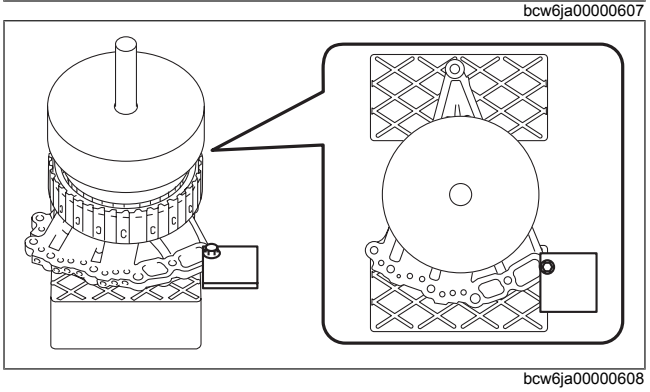
Caution
- **If the bolt and nut are tightened with excessive force when installing the SST, the alignment surface of the oil pump with the transaxle case could be damaged. Tighten the bolt and nut so that the SST does not move during low clutch clearance measurement.**

Note
- When installing the SST to the oil pump, use an M8 bolt and nut.



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SST installation bolt tightening torque
15 N·m {1.5 kgf·m, 11 ft·lbf} or less (tighten so that SST does not move during low clutch clearance measurement)

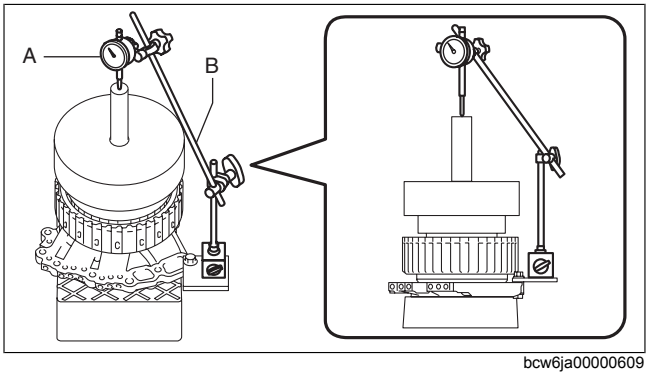

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1	SST (for securing magnetic stand)
2	Bolt (M8)
3	Nut (M8)

(2) Set the dial gauge and magnetic stand as shown in the figure.

Caution
- **To reduce measurement differences during the low clutch clearance measurement, set the dial gauge so that it is perpendicular to the alignment surface of the oil pump with the transaxle case.**

A : Dial gauge
B : Magnetic stand

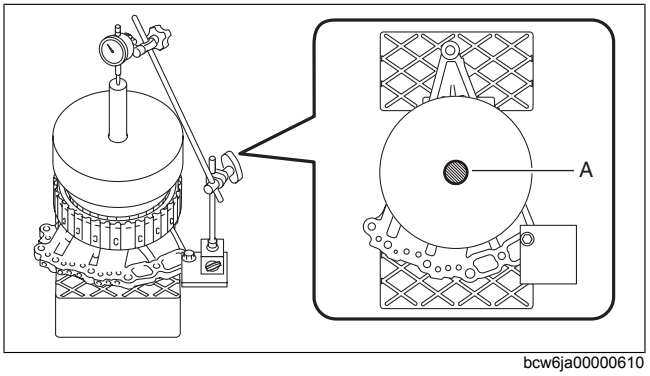


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(3) Set the dial gauge end near the center of the SST.

Caution
- **To reduce measurement differences during the low clutch clearance measurement, set the dial gauge end within the area shown in the figure.**

A : Dial gauge end set area



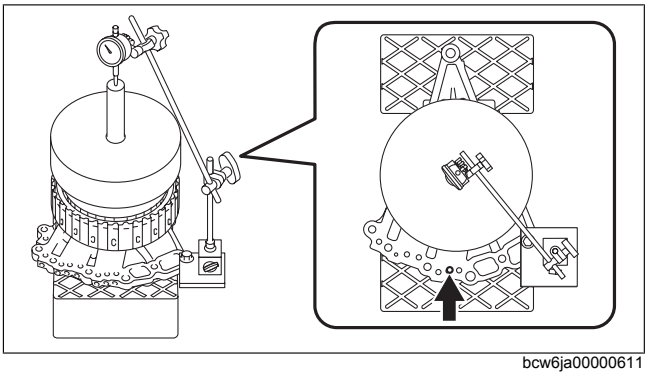
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7. Measure the low clutch clearance using the following procedure:
   (1) Blow compressed air into the oil passage shown in the figure to force the low clutch piston to stroke (approx. 3 times).

Warning
- **Always wear protective eye wear when using the air compressor. Otherwise, ATF or dirt particles blown off by the air compressor could get into the eyes.**

Caution
- **To prevent damage to parts, always use an air compressor which is adjusted to the indicated pressure.**

Compressed air pressure
0.39—0.44 MPa {4.0—4.4 kgf/cm2, 57—63 psi}



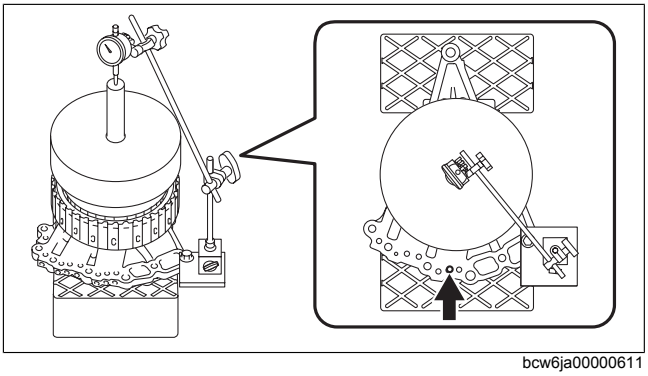
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(2) Blow compressed air into the oil passage shown in the figure to operate the low clutch piston, and read the value when the dial gauge is stabilized.

Warning
- **Always wear protective eye wear when using the air compressor. Otherwise, ATF or dirt particles blown off by the air compressor could get into the eyes.**

Caution
- **To prevent damage to parts, always use an air compressor which is adjusted to the indicated pressure.**
- **To reduce measurement differences during the low clutch clearance measurement, gently blow and release the compressed air.**

Compressed air pressure
0.39—0.44 MPa {4.0—4.4 kgf/cm2, 57—63 psi}



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(3) Input the dial gauge value, which was read while the low clutch piston was operating, into the measurement/adjustment value input sheet.
(4) Release the compressed air and read the dial gauge value while the low clutch piston is not operating.
(5) Input the dial gauge value, which was read while the low clutch piston was not operating, into the measurement/adjustment value input sheet.
(6) Perform the following calculation to calculate the low clutch clearance.
Low clutch clearance = B-C-A
A: Correction value for low clutch clearance (0.0440 mm {0.00173 in})
B: Dial gauge value with low clutch piston operated
C: Dial gauge value without low clutch piston operated

Note
Example
B: Dial gauge value with low clutch piston operated is 2.320 mm {0.09134 in}
C: Dial gauge value without low clutch piston operated is 0.595 mm {0.0234 in}
Low clutch clearance = 2.320 mm {0.09134 in} - 0.595 mm {0.0234 in} - 0.0440 mm {0.00173 in} = 1.6810 mm {0.066181 in}

Note
- Because a wave spring is included on the low clutch, a correction is necessary for the low clutch clearance based on the weight of the SST used during the low clutch clearance measurement. The correction value is 0.0440 mm {0.00173 in}.

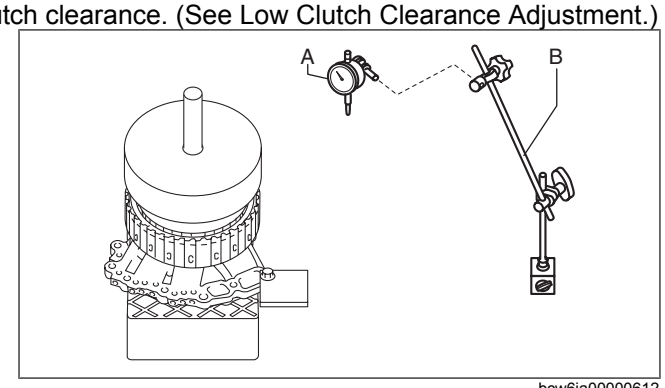
(7) Input the calculated low clutch clearance into the measurement/adjustment value input sheet.
(8) Verify that the low clutch clearance satisfies the specification.

Specification
1.330—1.530 mm {0.05237—0.06023 in}

- If not within the specification, adjust the low clutch clearance. (See Low Clutch Clearance Adjustment.)

8. Remove the dial gauge and magnetic stand.

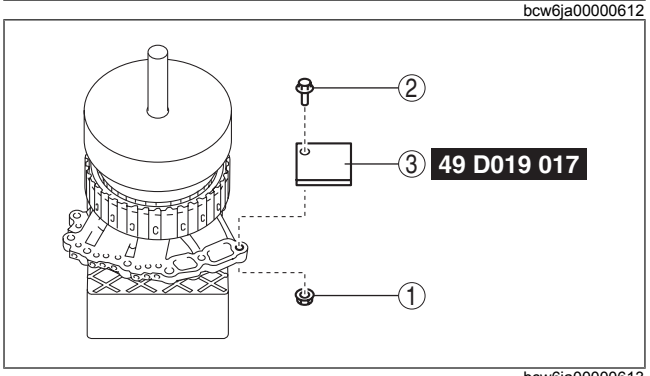
A : Dial gauge
B : Magnetic stand



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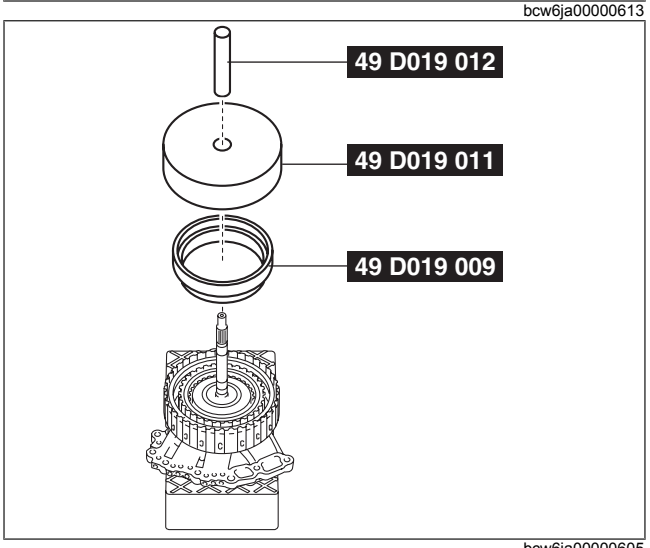
9. Remove the SST for securing the magnetic stand using the procedure shown in the figure.

1	Nut (M8)
2	Bolt (M8)
3	SST (for securing magnetic stand)



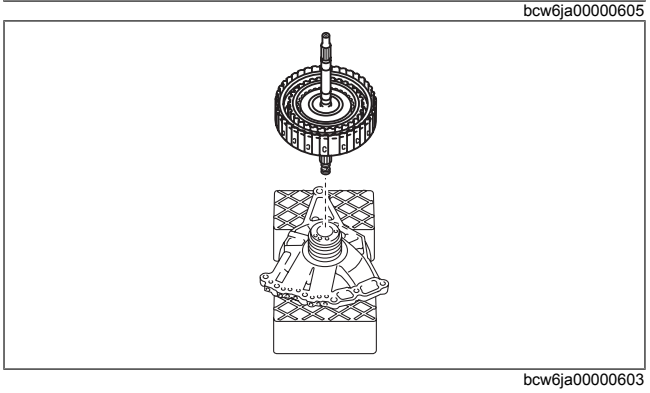
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10. Remove the SSTs.



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11. Remove the clutch component.

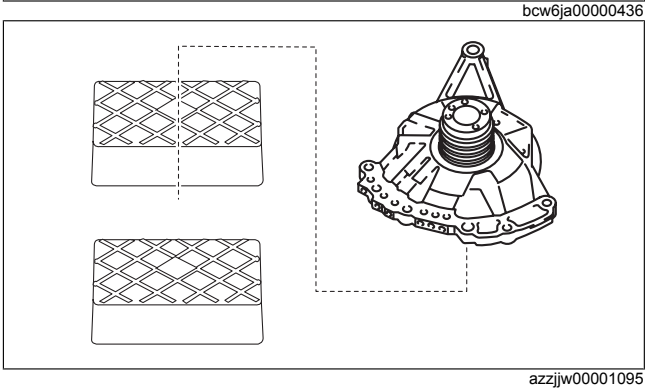

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12. Remove the thrust washer.



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13. Take the oil pump off the rubber plates.

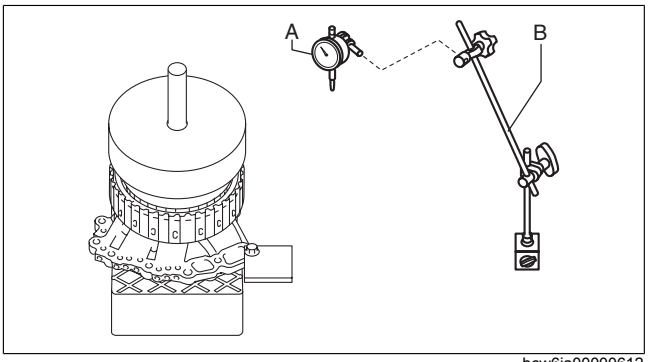


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Low Clutch Clearance Adjustment

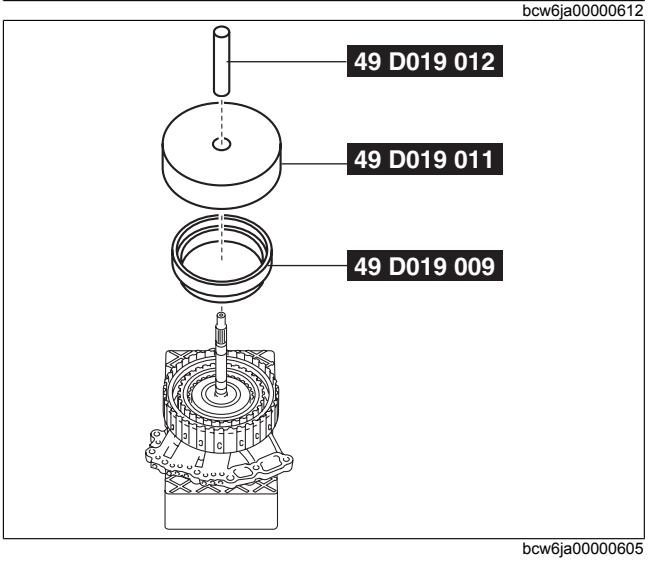
1. Remove the dial gauge and magnetic stand.

A : Dial gauge
B : Magnetic stand



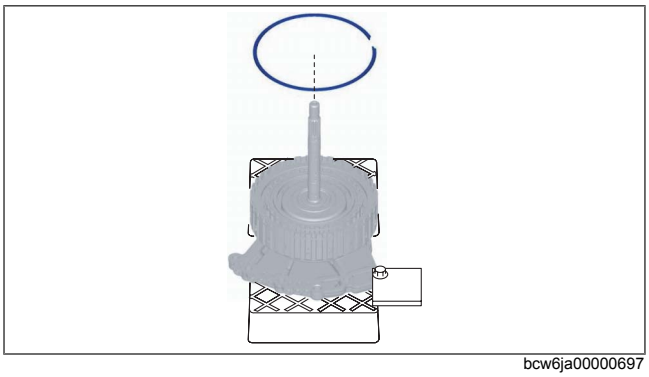
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2. Remove the SSTs.



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3. Remove the snap ring.
4. Measure the thickness of the removed snap ring.


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Note
- Recommended measuring instrument: Micrometer

5. Input the measured snap ring thickness into the measurement/adjustment value input sheet.
6. Perform the following calculation to calculate the low clutch clearance gap.

Note
- The low clutch clearance gap is the difference between the low clutch clearance and the median value of the low clutch clearance specification.

Low clutch clearance gap = D - H
D: Low clutch clearance
H: Median value of low clutch clearance specification (1.430 mm {0.05630 in})

Note
Example
D: Low clutch clearance is 1.6810 mm {0.066181 in}
Low clutch clearance gap = 1.6810 mm {0.066181 in} - 1.430 mm {0.05630 in} = 0.2510 mm {0.009882 in}

7. Input the calculated gap of the low clutch clearance into the measurement/adjustment value input sheet.
8. Perform the following calculation to calculate the optimum snap ring thickness.

Thickness of optimum snap ring = G + I
G: Thickness of removed snap ring
I: Low clutch clearance gap

Note
Example
G: Thickness of removed snap ring is 1.705 mm {0.06713 in}
Thickness of optimum snap ring = 1.705 mm {0.06713 in} + 0.2510 mm {0.009882 in} = 1.9560 mm {0.077008 in}

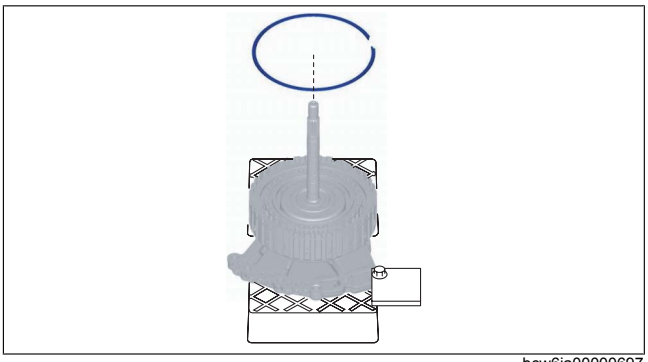
9. Input the calculated optimum snap ring thickness into the measurement/adjustment value input sheet.
10. From the following table, select a snap ring of a thickness closest to the calculated optimum snap ring thickness.

Snap ring thickness
2.2 mm {0.087 in}
2.1 mm {0.083 in}
2.0 mm {0.079 in}
1.9 mm {0.075 in}
1.8 mm {0.071 in}
1.7 mm {0.067 in}
1.6 mm {0.063 in}
1.5 mm {0.059 in}
1.4 mm {0.055 in}
1.3 mm {0.051 in}

11. Assemble the selected snap ring to the position shown in the figure.

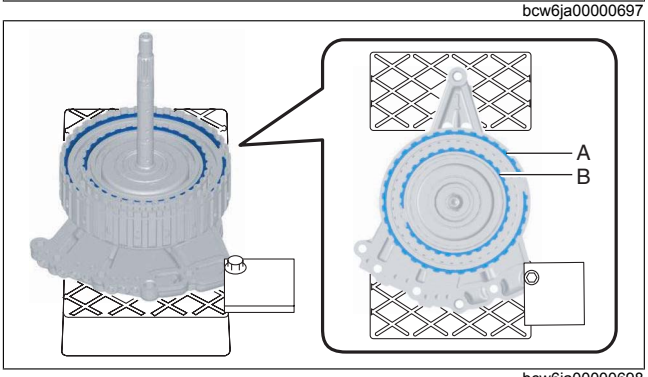
Caution
- **Assemble so that the end gap of the snap ring is positioned diagonally opposed to the end gap of the snap ring for the high clutch.**
- **After assembling the snap ring, verify that the snap ring is securely inserted into the bottom of the snap ring groove.**

Note
- Snap ring size: Outer diameter approx. 158.8 mm {6.252 in}



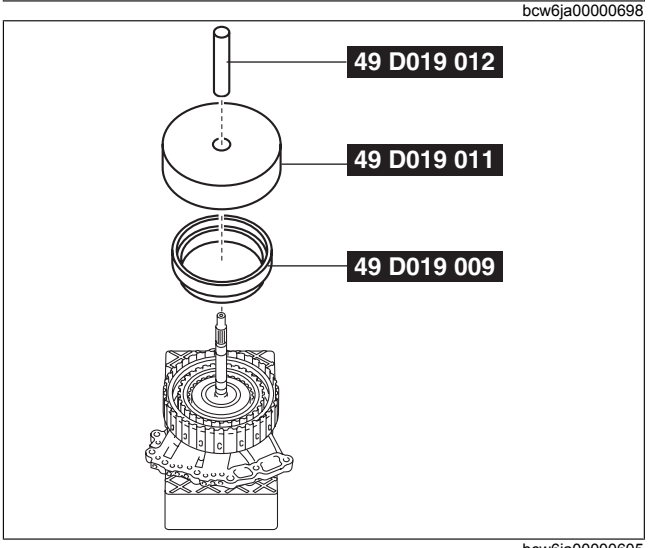
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A : Snap ring (low clutch)
B : Snap ring (high clutch)



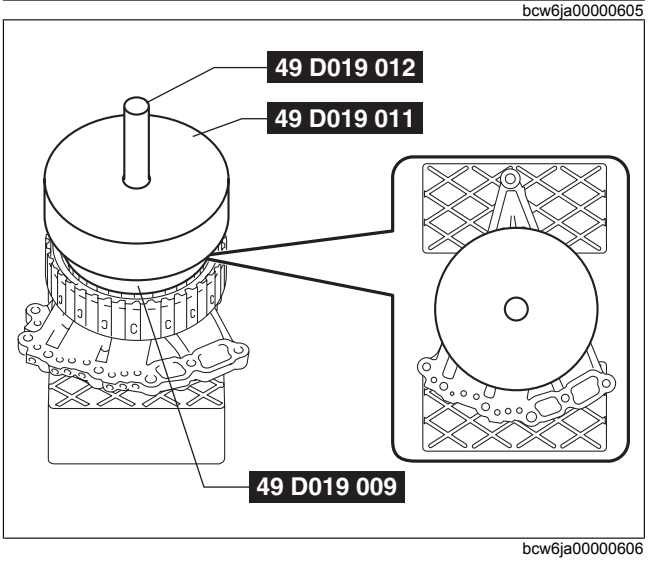
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12. Install the SSTs.



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13. Perform the low clutch clearance measurement from Step 6 (2). (See Low Clutch Clearance Measurement.)



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